TASK1

This task analyzes the impact of L1 data cache block size and associativity on miss rate and Instructions Per Cycle (IPC). The goal is to determine the optimal configuration (block size and associativity) that maximizes average IPC while adhering to a fixed 128 KB cache size. The configurations tested include block sizes of 32, 64, and 128 bytes, and associativities ranging from direct-mapped to 8-way.

INSTRUCTION PER CYCLE (IPC)							
BLOCK SIZE	ASSOCIATIVITY	benchmark1	benchmark2	benchmark3	benchmark4	benchmark5	average
32	direct-mapped	1.386871	1.518124	1.63251	0.075785	0.409966	1.0046512
32	2-way	1.432365	1.518981	1.639264	0.075797	0.409971	1.0152756
32	4-way	1.432686	1.519167	1.640427	0.075798	0.409976	1.0156108
32	8-way	1.43347	1.519405	1.641196	0.075813	0.410006	1.015978
64	direct-mapped	1.729832	1.862265	1.667956	0.134488	0.720932	1.2230946
64	2-way	1.786518	1.865193	1.683364	0.134541	0.721122	1.2381476
64	4-way	1.788074	1.86557	1.683216	0.134566	0.721107	1.2385066
64	8-way	1.789027	1.866904	1.683951	0.134568	0.721072	1.2391044
128	direct-mapped	1.85119	2.085583	1.678514	0.200997	1.070302	1.3773172
128	2-way	1.944334	2.090528	1.694094	0.201003	1.071007	1.4001932
128	4-way	1.947526	2.092548	1.695447	0.201082	1.071124	1.4015454
128	8-way	1.948042	2.092157	1.694365	0.201112	1.071007	1.4013366

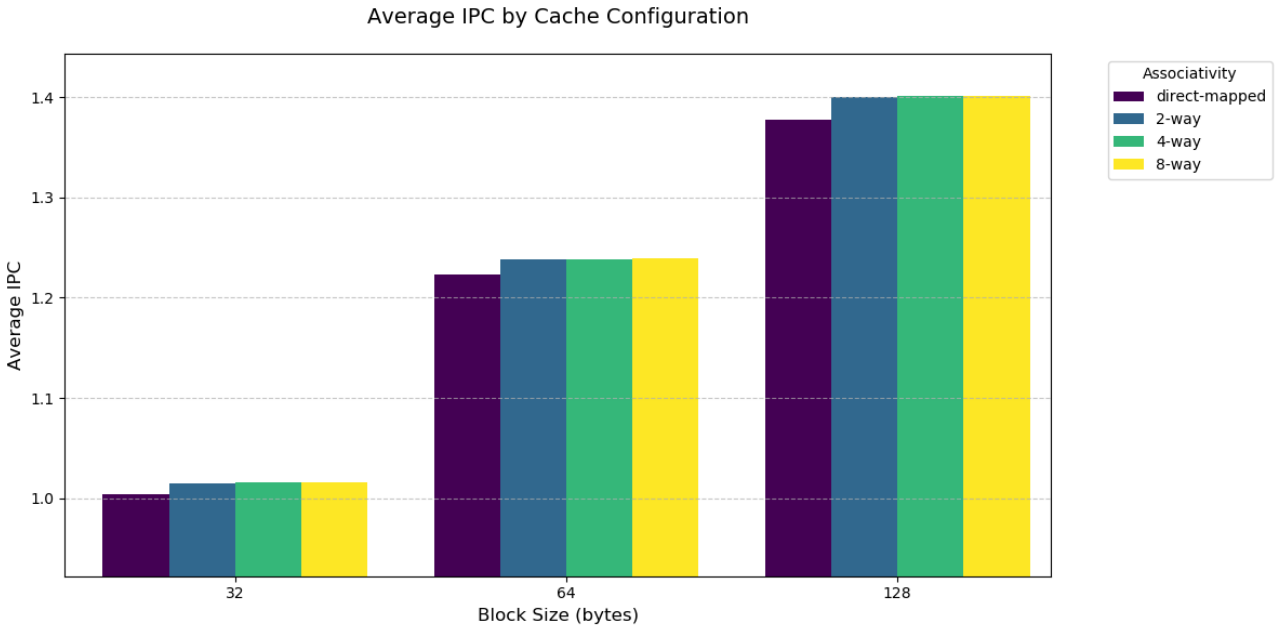
MISS-RATE							
BLOCK SIZE	ASSOCIATIVITY	benchmark1	benchmark2	benchmark3	benchmark4	benchmark5	average
32	direct-mapped	0.188218	0.006812	0.006458	0.199682	0.116635	0.103561
32	2-way	0.167158	0.006135	0.005792	0.199375	0.116384	0.0989688
32	4-way	0.166449	0.005894	0.005873	0.199299	0.116384	0.0987798
32	8-way	0.1664	0.005703	0.006055	0.199346	0.116384	0.0987776
64	direct-mapped	0.172298	0.00457	0.004234	0.100025	0.058605	0.0679464
64	2-way	0.149211	0.003919	0.003035	0.099647	0.058195	0.0628014
64	4-way	0.148314	0.00377	0.003072	0.099594	0.058195	0.062589
64	8-way	0.148336	0.003682	0.003155	0.099519	0.058195	0.0625774
128	direct-mapped	0.132874	0.003279	0.003559	0.050493	0.029941	0.0440292
128	2-way	0.108991	0.002526	0.001783	0.049961	0.029099	0.038472
128	4-way	0.108071	0.002392	0.001787	0.049838	0.029099	0.0382374
128	8-way	0.108053	0.002363	0.001833	0.049758	0.029099	0.0382212

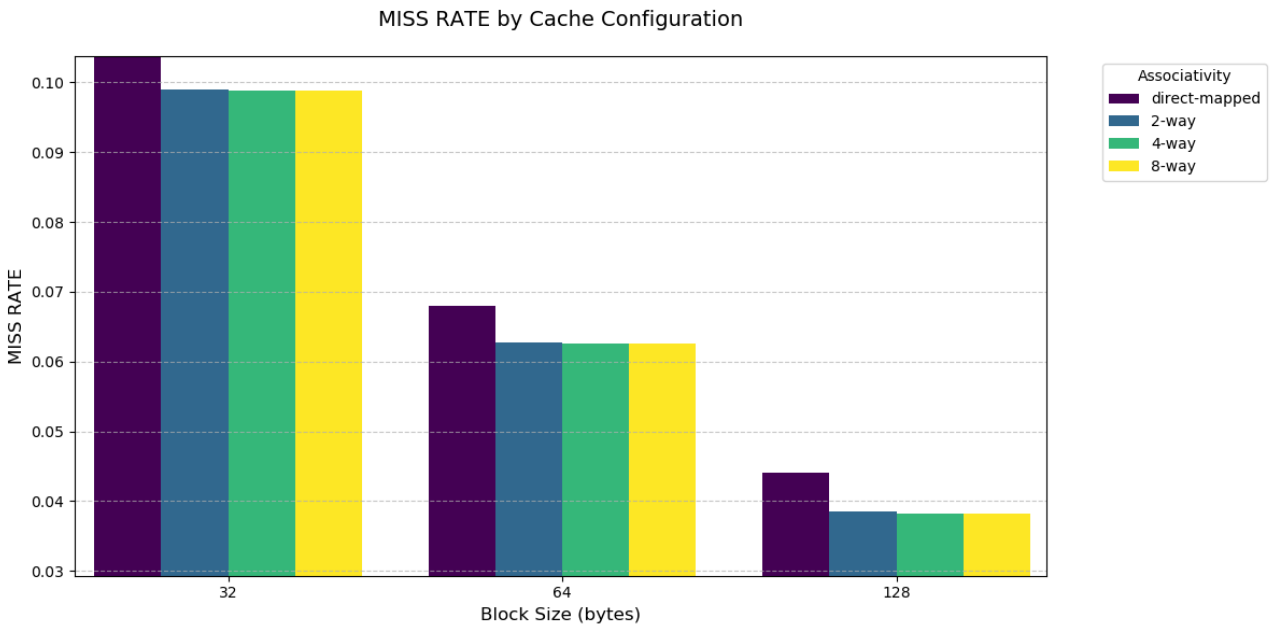
The experimental results reveal clear trends in the relationship between block size, associativity, and cache performance. Larger block sizes consistently improve both IPC and miss rates across all associativity levels. For instance, increasing the block size from 32B to 128B elevates the average IPC by approximately 39% (from 1.0047 to 1.4015)

while reducing the average miss rate by over 63% (from 0.1036 to 0.0382). This significant improvement stems from spatial locality exploitation: larger blocks fetch more contiguous

data per miss, reducing compulsory misses and amortizing memory access costs. Notably, benchmarks 1–3 benefit the most from this trend, as their access patterns likely exhibit strong spatial locality. Associativity also plays a critical role, though its impact diminishes at higher levels. For example, with a 128B block, upgrading from direct-mapped to 4-way associativity reduces the miss rate by 13% (from 0.0440 to 0.0382) and boosts IPC by 1.7% (from 1.3773 to 1.4015). However, increasing associativity beyond 4-way (e.g., to 8-way) yields negligible gains (IPC difference of 0.0002), suggesting that conflict misses are sufficiently mitigated at 4-way. Higher associativity introduces diminishing returns due to increased hardware complexity, access latency, and energy consumption. Benchmark-specific behavior highlights workload-dependent variations. Benchmarks 4 and 5 exhibit disproportionately low IPC values (e.g., 0.075–0.201 for 128B configurations), likely due to irregular or unpredictable memory access patterns that negate the benefits of spatial locality. These benchmarks suffer from frequent cache misses, underscoring the challenge of optimizing caches for diverse workloads.

It should be noted that, 8-way associativity further reduces the likelihood of conflict misses compared to 4-way, particularly in workloads with irregular access patterns or high contention for cache lines. This could prove advantageous in scenarios where L2 inclusivity policies (Task 2) or prefetchers (Task 3) introduce additional pressure on the memory hierarchy. In Task 2, optimizing the L2 cache (e.g., inclusivity policies, replacement algorithms) may amplify the benefits of a highly associative L1. For example, a more inclusive L2 could better support an 8-way L1 by reducing redundant fetches or coherence overheads, indirectly improving overall system efficiency. That's why for the rest of the tasks I chose the configuration of 128bytes of block size and 8-way associativity. Below there are bar graphs illustrating IPC and miss rate.




MISS RATE by Cache Configuration
MISS RATE
0.10
0.09
0.08
0.07
0.06
0.05
0.04
0.03
32
64
128
Block Size (bytes)
Associativity
direct-mapped
2-way
4-way
8-way

TASK2

- **IPC**

INSTRUCTION PER CYCLE (IPC)									
L2 CACHE SIZE	ASSOCIATIVITY	REPLACEMENT POLICY	INCLUSIVITY	benchmark1	benchmark2	benchmark3	benchmark4	benchmark5	average
512	4-way	FIFO	mostly_excl	1.862834	2.096085	1.705253	0.198534	1.072	1.3869412
512	4-way	FIFO	mostly_incl	2.297154	2.098191	1.706162	0.197779	1.051054	1.470068
512	4-way	RANDOM	mostly_excl	1.863708	2.095271	1.705253	0.182431	0.982405	1.3658136
512	4-way	RANDOM	mostly_incl	2.278749	2.095373	1.706152	0.177765	0.949212	1.4414502
512	4-way	LRU	mostly_excl	1.862834	2.096085	1.705253	0.198534	1.072	1.3869412
512	4-way	LRU	mostly_incl	2.302901	2.099858	1.706904	0.197866	1.051649	1.4718356
512	8-way	FIFO	mostly_excl	1.862605	2.095835	1.705253	0.198644	1.072	1.3868674
512	8-way	FIFO	mostly_incl	2.291696	2.099748	1.707059	0.197903	1.051399	1.469561
512	8-way	RANDOM	mostly_excl	1.865002	2.093491	1.705253	0.186757	1.003978	1.3708962
512	8-way	RANDOM	mostly_incl	2.269651	2.099387	1.706058	0.19118	1.018515	1.4569582
512	8-way	LRU	mostly_excl	1.862605	2.095835	1.705253	0.198644	1.072	1.3868674
512	8-way	LRU	mostly_incl	2.296452	2.100615	1.706969	0.19785	1.052364	1.47085
512	16-way	FIFO	mostly_excl	1.862884	2.095576	1.705253	0.198599	1.071998	1.386862
512	16-way	FIFO	mostly_incl	2.279725	2.099866	1.706934	0.197888	1.050987	1.46708
512	16-way	RANDOM	mostly_excl	1.867167	2.093496	1.705253	0.195434	1.04968	1.382206
512	16-way	RANDOM	mostly_incl	2.257973	2.100637	1.706953	0.19935	1.061458	1.4652742
512	16-way	LRU	mostly_excl	1.862884	2.095576	1.705253	0.198599	1.071998	1.386862
512	16-way	LRU	mostly_incl	2.285824	2.101303	1.706969	0.197861	1.05179	1.4687494
1024	4-way	FIFO	mostly_excl	1.871461	2.103645	1.705253	0.198437	1.072446	1.3902484
1024	4-way	FIFO	mostly_incl	2.326816	2.105604	1.706969	0.197829	1.051721	1.4777878
1024	4-way	RANDOM	mostly_excl	1.931461	2.100852	1.705253	0.18246	0.984049	1.380815
1024	4-way	RANDOM	mostly_incl	2.3124	2.103784	1.706969	0.178183	0.950354	1.450338
1024	4-way	LRU	mostly_excl	1.871461	2.103645	1.705253	0.198437	1.072446	1.3902484
1024	4-way	LRU	mostly_incl	2.331628	2.107798	1.706969	0.197907	1.051476	1.4791556
1024	8-way	FIFO	mostly_excl	1.871423	2.103349	1.705253	0.19854	1.072446	1.3902022
1024	8-way	FIFO	mostly_incl	2.32558	2.10559	1.706969	0.197885	1.051721	1.477549
1024	8-way	RANDOM	mostly_excl	1.935084	2.099944	1.705253	0.170931	0.926022	1.3674468
1024	8-way	RANDOM	mostly_incl	2.310751	2.101557	1.706969	0.167391	0.892768	1.4358872
1024	8-way	LRU	mostly_excl	1.871423	2.103349	1.705253	0.19854	1.072446	1.3902022
1024	8-way	LRU	mostly_incl	2.331545	2.107451	1.706969	0.197862	1.053055	1.4793764
1024	16-way	FIFO	mostly_excl	1.870559	2.103729	1.705253	0.198713	1.072446	1.39014
1024	16-way	FIFO	mostly_incl	2.326382	2.106882	1.706969	0.197919	1.052353	1.478101
1024	16-way	RANDOM	mostly_excl	1.936221	2.099373	1.705253	0.182207	0.98194	1.3809988
1024	16-way	RANDOM	mostly_incl	2.313124	2.106066	1.706969	0.186191	0.993748	1.4612196
1024	16-way	LRU	mostly_excl	1.870559	2.103729	1.705253	0.198713	1.072446	1.39014
1024	16-way	LRU	mostly_incl	2.332178	2.108836	1.706969	0.197934	1.052194	1.4796222
2048	4-way	FIFO	mostly_excl	2.361035	2.11134	1.705253	0.198607	1.073252	1.4898974
2048	4-way	FIFO	mostly_incl	2.360484	2.115517	1.706969	0.19792	1.053814	1.4869408
2048	4-way	RANDOM	mostly_excl	2.352869	2.109051	1.705253	0.182551	0.98536	1.4670168
2048	4-way	RANDOM	mostly_incl	2.358985	2.115593	1.706969	0.17812	0.953235	1.4625804
2048	4-way	LRU	mostly_excl	2.361035	2.11134	1.705253	0.198607	1.073252	1.4898974
2048	4-way	LRU	mostly_incl	2.360516	2.117959	1.706969	0.197888	1.052756	1.4872176
2048	8-way	FIFO	mostly_excl	2.360907	2.111687	1.705253	0.19857	1.073252	1.4899338
2048	8-way	FIFO	mostly_incl	2.360076	2.11482	1.706969	0.19794	1.052756	1.4865122
2048	8-way	RANDOM	mostly_excl	2.352292	2.109863	1.705253	0.171288	0.929632	1.4536656
2048	8-way	RANDOM	mostly_incl	2.358527	2.115253	1.706969	0.167874	0.898131	1.4493508
2048	8-way	LRU	mostly_excl	2.360907	2.111687	1.705253	0.19857	1.073252	1.4899338
2048	8-way	LRU	mostly_incl	2.359981	2.118488	1.706969	0.197898	1.052756	1.4872184
2048	16-way	FIFO	mostly_excl	2.360961	2.111687	1.705253	0.198676	1.073252	1.4899658
2048	16-way	FIFO	mostly_incl	2.359237	2.115208	1.706969	0.197881	1.052756	1.4864102
2048	16-way	RANDOM	mostly_excl	2.353645	2.109222	1.705253	0.16237	0.885857	1.4432694
2048	16-way	RANDOM	mostly_incl	2.358333	2.114716	1.706969	0.157676	0.859802	1.4394992
2048	16-way	LRU	mostly_excl	2.360961	2.111687	1.705253	0.198676	1.073252	1.4899658
2048	16-way	LRU	mostly_incl	2.35908	2.118138	1.706969	0.197896	1.052756	1.4869678
								MAX IPC:	1.4899658

The experimental results for Task 2 highlight the interplay between L2 cache size, associativity, replacement policies, and inclusivity in optimizing IPC. Increasing the L2 cache size from 512KB to 2048KB consistently improves performance, with the 2048KB configurations achieving the highest average IPC (1.4899). Larger caches reduce capacity

misses by accommodating larger working sets, particularly benefiting benchmarks 1–3, which exhibit strong spatial and temporal locality. For instance, Benchmark1's IPC improves by ~25% (from 1.948 in Task 1 to 2.361 in Task 2) under optimal configurations. Replacement policies significantly influence performance. LRU and FIFO outperform RandomRP across all cache sizes, as they better exploit temporal locality. RandomRP's unpredictable evictions lead to higher miss rates, undermining performance in workloads with recurring access patterns.

Inclusivity also plays a big role. The mostly_excl policy marginally outperforms mostly_incl for larger caches (e.g., 2048KB), as it avoids redundant storage of L1 data in the L2, freeing space for unique blocks. However, with smaller caches (512KB), mostly_incl occasionally performs better (e.g., 1.4718 IPC), likely due to improved coherence management in constrained cache budgets. Associativity exhibits diminishing returns. While higher associativity (16-way) slightly reduces conflict misses, it offers negligible gains over 4-way or 8-way associativity for larger caches. Benchmarks 4 and 5 remain performance outliers, with IPC values stagnating near 0.16–0.20, indicating persistent inefficiencies in their memory access patterns. These benchmarks likely suffer from irregular or scattered accesses that evade cache optimizations, emphasizing the challenge of designing universally effective cache hierarchies. The best configuration for this task (highlighted in green) is either FIFO or LRU replacement policy for 2048kb of L2 cache size and 16-way associativity. I chose LRU since it looks like it performs slightly better regarding the miss rate as shown and described below.

- **MISS RATE**

The miss rate data reveals critical insights into how L2 cache configurations influence memory efficiency. Larger L2 cache sizes (2048KB) consistently achieve the lowest average miss rates (~0.025), significantly outperforming smaller caches (512KB: ~0.038). This improvement stems from reduced capacity misses, as larger caches accommodate larger working sets, particularly benefiting benchmarks 1–3. For instance, Benchmark1's miss rate drops from 0.110 (512KB) to 0.043 (2048KB) under optimal configurations.

Replacement policies strongly impact miss rates. LRU and FIFO outperform RandomRP, with RandomRP configurations suffering up to 30% higher miss rates (e.g., 0.025 vs. 0.038 for 2048KB). LRU/FIFO's reliance on temporal locality ensures high-priority data remains cached, while RandomRP's arbitrary evictions increase misses.

Inclusivity exhibits nuanced trade-offs. The mostly_incl policy slightly reduces miss rates in smaller caches (512KB: ~0.025 vs. ~0.038 for mostly_excl) by maintaining coherence between L1 and L2. However, for larger caches (2048KB), mostly_excl achieves comparable miss rates (~0.025) while avoiding redundancy, freeing L2 space for unique blocks.

Associativity has minimal impact on miss rates for large caches. For example, 2048KB with 4-way associativity achieves nearly identical miss rates (~0.025) as 16-way, suggesting that conflict misses are negligible when cache capacity is sufficient. Smaller caches (512KB) see marginal gains with higher associativity (e.g., 16-way: 0.027 vs. 4-

way: 0.038), but these improvements are overshadowed by the dominance of capacity misses. Best configuration for miss rate (higlighted in blue) is actually 1024 cache size, 16-way associativity with LRU replacement policy and mostly inclusive. While this seems counterintuitive compared to larger caches (e.g., 2048KB), the result stems from the interplay of associativity, inclusivity, and workload-specific behavior

MISS-RATE										
L2 CACHE SIZE	ASSOCIATIVITY	REPLACEMENT POLICY	INCLUSIVITY	benchmark1	benchmark2	benchmark3	benchmark4	benchmark5	average	IPC/MISS_RATE
512	4-way	FIFO	mostly_excl	0.110067	0.0024	0.001655	0.049776	0.029099	0.0385994	35.9316776944719
512	4-way	FIFO	mostly_incl	0.045983	0.002309	0.001648	0.049748	0.029099	0.0257574	57.073617678803
512	4-way	RANDOM	mostly_excl	0.108685	0.002403	0.001655	0.049767	0.029099	0.0383218	35.6406431848191
512	4-way	RANDOM	mostly_incl	0.045857	0.002312	0.001648	0.04975	0.029099	0.0257332	56.0151943792455
512	4-way	LRU	mostly_excl	0.110067	0.0024	0.001655	0.049776	0.029099	0.0385994	35.9316776944719
512	4-way	LRU	mostly_incl	0.045703	0.00229	0.001648	0.049742	0.029099	0.0256964	57.2778910664529
512	8-way	FIFO	mostly_excl	0.110063	0.0024	0.001655	0.049767	0.029099	0.0385968	35.9321860879658
512	8-way	FIFO	mostly_incl	0.048152	0.002309	0.001648	0.049752	0.029099	0.026192	56.1072464874771
512	8-way	RANDOM	mostly_excl	0.108525	0.002404	0.001655	0.049767	0.029099	0.03829	35.8029825019587
512	8-way	RANDOM	mostly_incl	0.047804	0.002313	0.001648	0.049742	0.029099	0.0261212	55.7768479242914
512	8-way	LRU	mostly_excl	0.110063	0.0024	0.001655	0.049767	0.029099	0.0385968	35.9321860879658
512	8-way	LRU	mostly_incl	0.047932	0.002289	0.001648	0.049749	0.029099	0.0261434	56.2608535997613
512	16-way	FIFO	mostly_excl	0.110066	0.0024	0.001655	0.049772	0.029099	0.0385984	35.9305567070138
512	16-way	FIFO	mostly_incl	0.052432	0.002309	0.001648	0.049751	0.029099	0.0270478	54.2402709277649
512	16-way	RANDOM	mostly_excl	0.108271	0.002404	0.001655	0.049773	0.029099	0.0382404	36.1451763056872
512	16-way	RANDOM	mostly_incl	0.050426	0.00231	0.001648	0.049753	0.029099	0.0266472	54.9879236842895
512	16-way	LRU	mostly_excl	0.110066	0.0024	0.001655	0.049772	0.029099	0.0385984	35.9305567070138
512	16-way	LRU	mostly_incl	0.052347	0.002289	0.001648	0.049738	0.029099	0.0270242	54.3494127485735
1024	4-way	FIFO	mostly_excl	0.110056	0.002401	0.001655	0.049773	0.029099	0.0385968	36.019784023546
1024	4-way	FIFO	mostly_incl	0.043266	0.002301	0.001648	0.049743	0.029099	0.0252114	58.6158563189668
1024	4-way	RANDOM	mostly_excl	0.094392	0.002402	0.001655	0.049774	0.029099	0.0354644	38.9352421019389
1024	4-way	RANDOM	mostly_incl	0.043267	0.002301	0.001648	0.049754	0.029099	0.0252138	57.5215953168503
1024	4-way	LRU	mostly_excl	0.110056	0.002401	0.001655	0.049773	0.029099	0.0385968	36.019784023546
1024	4-way	LRU	mostly_incl	0.042876	0.002289	0.001648	0.049748	0.029099	0.025132	58.8554671335349
1024	8-way	FIFO	mostly_excl	0.110057	0.002401	0.001655	0.049769	0.029099	0.0385962	36.0191469626544
1024	8-way	FIFO	mostly_incl	0.043101	0.0023	0.001648	0.049748	0.029099	0.0251792	58.6813322107136
1024	8-way	RANDOM	mostly_excl	0.093076	0.002404	0.001655	0.04977	0.029099	0.0352008	38.847037567328
1024	8-way	RANDOM	mostly_incl	0.043415	0.0023	0.001648	0.049737	0.029099	0.0252398	56.8898010285343
1024	8-way	LRU	mostly_excl	0.110057	0.002401	0.001655	0.049769	0.029099	0.0385962	36.0191469626544
1024	8-way	LRU	mostly_incl	0.042856	0.002289	0.001648	0.049751	0.029099	0.0251286	58.8722173141361
1024	16-way	FIFO	mostly_excl	0.110059	0.0024	0.001655	0.049765	0.029099	0.0385956	36.0180953269285
1024	16-way	FIFO	mostly_incl	0.043105	0.0023	0.001648	0.049753	0.029099	0.025181	58.6990588141853
1024	16-way	RANDOM	mostly_excl	0.093304	0.002403	0.001655	0.049763	0.029099	0.0352448	39.1830511167605
1024	16-way	RANDOM	mostly_incl	0.043462	0.002298	0.001648	0.049744	0.029099	0.0252502	57.8696247950511
1024	16-way	LRU	mostly_excl	0.110059	0.0024	0.001655	0.049765	0.029099	0.0385956	36.0180953269285
1024	16-way	LRU	mostly_incl	0.042851	0.002289	0.001648	0.049745	0.029099	0.0251264	58.8871545466123
2048	4-way	FIFO	mostly_excl	0.043037	0.0024	0.001655	0.04977	0.029099	0.0251922	59.141218313605
2048	4-way	FIFO	mostly_incl	0.043048	0.002293	0.001648	0.049748	0.029099	0.0251672	59.0824883181284
2048	4-way	RANDOM	mostly_excl	0.043289	0.0024	0.001655	0.04978	0.029099	0.0252446	58.1121031824628
2048	4-way	RANDOM	mostly_incl	0.043043	0.002295	0.001648	0.049747	0.029099	0.0251664	58.1163932862865
2048	4-way	LRU	mostly_excl	0.043037	0.0024	0.001655	0.04977	0.029099	0.0251922	59.141218313605
2048	4-way	LRU	mostly_incl	0.043039	0.002289	0.001648	0.049748	0.029099	0.0251646	59.0995922843995
2048	8-way	FIFO	mostly_excl	0.043036	0.002401	0.001655	0.04977	0.029099	0.0251922	59.1426632052778
2048	8-way	FIFO	mostly_incl	0.043041	0.002294	0.001648	0.049756	0.029099	0.0251676	59.0645194615299
2048	8-way	RANDOM	mostly_excl	0.043463	0.002404	0.001655	0.049764	0.029099	0.025277	57.5094196304941
2048	8-way	RANDOM	mostly_incl	0.043041	0.002293	0.001648	0.049754	0.029099	0.025167	57.5893352405928
2048	8-way	LRU	mostly_excl	0.043036	0.002401	0.001655	0.04977	0.029099	0.0251922	59.1426632052778
2048	8-way	LRU	mostly_incl	0.043026	0.002289	0.001648	0.049747	0.029099	0.0251618	59.1062006692685
2048	16-way	FIFO	mostly_excl	0.043036	0.0024	0.001655	0.049773	0.029099	0.0251926	59.142994371363
2048	16-way	FIFO	mostly_incl	0.043038	0.002294	0.001648	0.049744	0.029099	0.0251646	59.0675075304197
2048	16-way	RANDOM	mostly_excl	0.043468	0.002402	0.001655	0.049768	0.029099	0.0252784	57.094966453573
2048	16-way	RANDOM	mostly_incl	0.043049	0.002293	0.001648	0.049747	0.029099	0.0251672	57.1974315776089
2048	16-way	LRU	mostly_excl	0.043036	0.0024	0.001655	0.049773	0.029099	0.0251926	59.142994371363
2048	16-way	LRU	mostly_incl	0.043007	0.002289	0.001648	0.049746	0.029099	0.0251578	59.1056372178807
								MIN MR:	0.0251264	

TASK3

INSTRUCTION PER CYCLE (IPC)								
PREFETCHER	DEGREE	benchmark1	benchmark2	benchmark3	benchmark4	benchmark5	average	
Stride	1	2.379857	2.1315	1.705255	0.233517	1.238668	1.5377594	
Stride	2	2.388514	2.142067	1.705482	0.261006	1.341695	1.5677528	
Stride	3	2.383697	2.141069	1.705791	0.246971	1.270267	1.549559	
Stride	4	2.38711	2.144364	1.705632	0.248993	1.270267	1.5512732	
Tagged	1	2.390955	2.156354	1.706456	0.264744	1.430058	1.5897134	
Tagged	2	2.392758	2.169552	1.706196	0.291682	1.559525	1.6239426	
Tagged	3	2.396235	2.178417	1.706403	0.321023	1.681653	1.6567462	
Tagged	4	2.394911	2.180586	1.706312	0.3241	1.697262	1.6606342	

BANDWIDTH								
PREFETCHER	DEGREE	benchmark1	benchmark2	benchmark3	benchmark4	benchmark5	average	bandwidth/IPC
Stride	1	52818346	212879864	9800443	2509847444	2405778715	1038224962	675154359.258
Stride	2	53016605	214061344	9797379	2805277866	2606032262	1137637091	725648259.853
Stride	3	52934094	213906793	9803520	2654460069	2467197959	1079660487	696753390.481
Stride	4	52933487	214288185	9806974	2676176181	2467197959	1084080557	698832776.329
Tagged	1	54601004	223115384	10231090	2849397699	2777848787	1183038793	744183695.501
Tagged	2	57711015	229671215	10620455	3143099966	3029695235	1294159577	796924458.537
Tagged	3	60720969	234641674	10962481	3462362026	3267244792	1407186388	849367506.26
Tagged	4	64038010	238073960	11269852	3498035521	3297821144	1421847697	856207644.887

Based on the simulation data, the Tagged prefetcher with a degree of 4 achieves the best performance in terms of IPC, with an average of 1.6606, the highest among all configurations. This indicates that the Tagged prefetcher is generally more effective than the Stride prefetcher in improving instruction throughput. Additionally, as the prefetch degree increases, there is a consistent improvement in IPC, demonstrating that a higher prefetching degree results in better instruction-level parallelism.

In terms of bandwidth usage, the Tagged prefetcher with a degree of 4 also exhibits the highest memory bandwidth consumption, averaging 1.4218×10^9 bytes per second. This shows that while it maximizes IPC, it also exerts significant pressure on memory resources. The bandwidth per IPC ratio (856.2 million bytes per IPC) further highlights this trade-off, as it represents the highest ratio among all configurations.

To find the best balance between IPC and memory bandwidth, we need to consider both high IPC and reasonable bandwidth efficiency. The Tagged prefetcher with a degree of 3 provides a strong compromise, with an IPC of 1.6567 (very close to the highest value) and an average bandwidth of 1.4072×10^9 bytes per cycle, resulting in a slightly lower bandwidth per IPC ratio of 849.3 million. This suggests that it achieves almost the same computational efficiency while consuming slightly less memory bandwidth compared to the degree-4 configuration. Comparing to the Stride prefetcher, the Tagged prefetcher

outperforms it in both IPC and bandwidth efficiency. The best Stride configuration (degree 2) achieves an IPC of 1.5677 with a bandwidth per IPC ratio of 725.6 million, indicating that while it is more bandwidth-efficient, it does not maximize IPC as effectively as the Tagged prefetcher.

Thus, based on the trade-off between computational efficiency and memory traffic, the Tagged prefetcher with a degree of 3 is the optimal configuration. It provides near-optimal IPC while maintaining a slightly lower memory bandwidth demand than the maximum degree configuration, making it the best-balanced choice for system performance. Below I'm providing bar graphs visualizing IPC and Bandwidth for each configuration.

